Neuro-symbolic artificial intelligence

What is Neuro-symbolic AI?

Neuro-symbolic artificial intelligence combines neural networks with symbolic AI to enable systems to mimic human decision-making and logic. Neural networks excel at pattern recognition, while symbolic AI excels at logic, language, and rule-based reasoning, much like human comprehension. By combining these methods, AI can learn from data while applying logical reasoning to interpret and explain conclusions. By combining these methods, AI can learn from data while applying logical reasoning to interpret and explain conclusions.

- The major research areas within neuro-symbolic AI include learning and inference, logic and reasoning, and knowledge representation.
- Neuro-symbolic AI benefits regulated, high-impact fields where trust and accountability are essential, such as law, medicine, and safety-focused industries.
- You can use neuro-symbolic AI to support efficient, transparent decision-making in tasks such as legal document review, personalized patient care, and crisis management planning. For example, neuro-symbolic AI might identify objects in an image using pattern recognition from neural network algorithm components, then use symbolic logic to infer relationships between them. This enables deeper understanding and reasoning for applications, such as natural language processing, complex problem-solving, and decision-making.

Main Architectural Paradigms

- Neural → Symbolic: Neural networks interpret perceptual data (like images or speech) and extract clean symbols, which are then processed by a logical reasoner (e.g., Neural-Concept Learner).
- Symbolic → Neural: Symbolic systems generate or label structured training data to guide deep learning models (e.g., using automated provers).
- Deep Integration: Architectures like Logic Tensor Networks or DeepProbLog embed logical reasoning directly within a differentiable framework, enabling joint learning and inference.

Neuro-symbolic AI examples

Industries are just scratching the surface of potential neuro-symbolic applications. Thanks to the combined capabilities of neuro-symbolic systems, several industries can use this algorithm type to support efficient, transparent decision-making. Some examples of use cases include:

- Legal document analysis

Neuro-symbolic systems can interpret contracts and use rule-based reasoning to identify relevant clauses, assess compliance, and provide legal justifications with traceable decision paths.

- Crisis management planning

These systems can create strategic responses by integrating current data with symbolic models of crisis scenarios to make a formal decision. Neural networks can make sense of chaotic data, while symbolic AI can explain the reasoning behind recommended actions.

- Job candidate profiling

Whereas traditional AI systems have shown bias toward particular candidate profiles, neuro-symbolic systems can build a profile for candidates using thoughtful selection criteria and reasoning, providing transparent reasoning for hiring decisions. This increases the trust and fairness of the hiring process across organizations.

- Personalized medicine

Neuro-symbolic systems can use pre-established guidelines and deep learning to accurately analyze medical images, provide diagnoses, recommend optimal treatments for patients, and improve preventative care measures.